Qwen vs DeepSeek Latency

Beacon demo agent. Same 15 visitor messages and 20 qualification transcripts, sent through Beacon's own conversation code to each model. Measured 23 October 2026 on the development laptop.

DeepSeek deepseek-flash (hosted API) Qwen3-4B + Beacon adapter (open source, local GPU)

PRODUCT QUESTION, MEDIAN

3.8 s · 10.4 s

Qwen takes 2.8× as long. Time until the reply text is ready.

ASSIGNMENT BUDGET

1.5 s

Median, end of speech to first audio. Neither model meets it for product questions.

LEAD QUALIFICATION, MEDIAN

1.8 s · 7.5 s

Passed Beacon's schema check: DeepSeek 8/20, Qwen 20/20. Runs in the background, so the visitor doesn't wait for it.

MODEL COST PER PRODUCT QUESTION

$0.0016 · $0

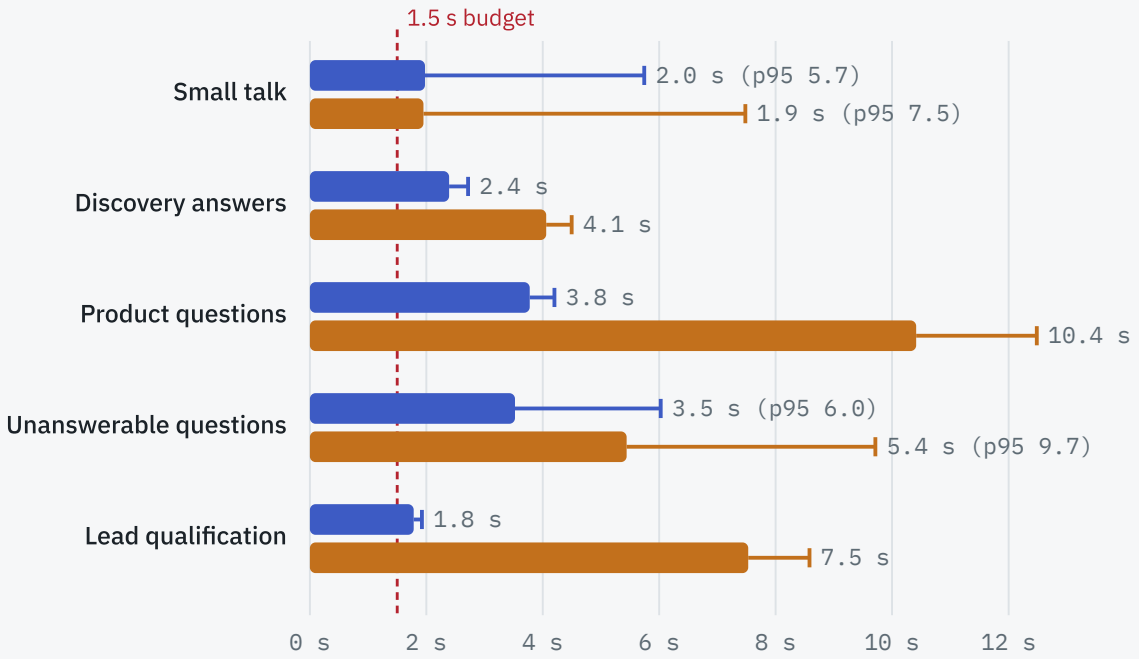
DeepSeek at peak price (repo cost model). Qwen has no per-call charge; it needs a GPU server.

Response time by message type

MESSAGE TYPE	DEEPSEEK P50	P95	QWEN P50	P95	QWEN ÷ DEEPSEEK
All conversation turns all 14 measured turns	3.3 s	5.7 s	7.5 s	11.7 s	2.3×
Small talk “hi”, “what can you do”	2.0 s	5.7 s	1.9 s	7.5 s	1.0×
Discovery answers “300”, “manually, in excel”	2.4 s	2.7 s	4.1 s	4.5 s	1.7×
Product questions “how to change status”, “How do I add units…”	3.8 s	4.2 s	10.4 s	12.5 s	2.8×
Unanswerable questions “How much does Leadrat cost?”	3.5 s	6.0 s	5.4 s	9.7 s	1.5×
Decision ready (plan) Beacon knows what to do and which screen to open	1.2 s	2.4 s	2.4 s	4.1 s	2.0×
Lead qualification 20 evaluation conversations; runs in the background	1.8 s	1.9 s	7.5 s	8.6 s	4.2×

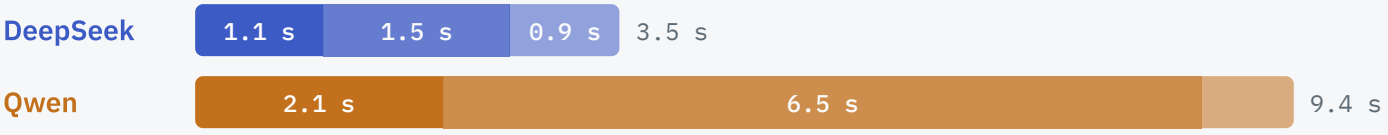
p50 is the median; p95 is the slowest typical case. Times run from the moment the message reaches the server until the reply text has passed its source check. The first turn of each run warmed the connection and prompt cache and is left out.

Median time, with p95



Bars show the median; the thin line runs to p95. The dashed line is the assignment's 1.5 s median budget for the whole voice turn.

Where a product question spends its time



Three model calls per product question: **plan** (what the visitor wants, which screen), **answer** (written from handbook passages) and **check** (every claim checked against those passages). Medians of each call. On Qwen the answer call dominates: generation runs at about 29 tokens a second on the 4 GB laptop GPU.

What this covers

- Measured: the language-model part of each turn, through Beacon's real conversation engine and handbook search. DeepSeek over the internet from India with a warm connection; Qwen3-4B-Instruct-2507 (Q4_K_M) with the trained beacon-qwen3-17k adapter on llama.cpp, all layers on an RTX 2050 (4 GB).
- Not measured: speech recognition (Soniox), voice synthesis and the CRM screen. These add to the voice turn; the cost doc reports 45–77 ms from checked text to first audio when the voice is prepared during the check.
- Qwen numbers depend on the GPU. This laptop GPU is entry level; a data-centre GPU would generate faster. That was not measured.
- Turns that ended in an error, which Beacon answers with a fallback: DeepSeek none; Qwen "i am a developer".
- Qualification quality differs from speed. DeepSeek wrote tools such as "Zoho CRM" as plain text where the schema expects a list, so 12 of 20 results were rejected. In the live app that means one retry, then the rules-based fallback that sends the lead to human review. The fine-tuned Qwen adapter passed all 20. Field accuracy was not checked here.
- > Assignment check: Qwen3-4B is openly licensed (Apache 2.0), but it has about 4 billion parameters. The assignment limits the fine-tuned base model to 3 billion.

▼ Every measured message

MESSAGE	TYPE	DEEPSEEK	QWEN	DEEPSEEK NOTE	QWEN NOTE
show leads	product	3.7 s	12.0 s		
how to change status	product	3.2 s	7.6 s		
How do I add units to a project?	product	3.8 s	10.8 s		
How do I import a spreadsheet of customer data?	product	3.3 s	11.7 s		
How does clock in work?	product	4.2 s	12.5 s		
lead ka status kaise badle	product	3.9 s	9.5 s		
lead me note kaise add kare	product	4.0 s	10.4 s		
How much does Leadrat cost?	refusal	6.0 s	5.4 s		
Does Leadrat integrate with Salesforce?	refusal	3.5 s	9.7 s		
hi	chat	2.0 s	1.9 s		
what can you do	chat	5.7 s	7.5 s		
i am a developer	discovery	2.2 s	3.6 s		error: ValueError
300	discovery	2.4 s	4.1 s		
manually, in excel	discovery	1.7 s	3.2 s		
around 2000 leads a month	discovery	2.7 s	4.5 s		